



Data Science Lifecycle

Roles and Responsibilities - Draft

DS Lifecycle Framework: Roles and Responsibilities

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

Stage Owners

The Stage owner will be responsible for implementing the DS Lifecycle and making sure the soft reviews are happening for each step in the Stage. They should be the (or at least one of the) accountable owners in the QA and stage gate sign-off steps. Example roles could be the use case product owner or a DS Lead/HoD.

Step Owners

Responsible Owner

Role

The responsible owner for a step should be close to the solution design and implementation activities such as a Data Scientist, MLE.

Responsibilities

Responsible owners are responsible ensuring the checklists have been populated. They are not necessarily directly responsible for filling in the documentations but are responsible for working with relevant stakeholders to provide evidence. Should an item on the checklist not be applicable to the use case or can be deprioritised, a justification must be provided instead.

Accountable Owner

Role

The accountable owner is should be at least one level senior than responsible owner for that step. For QA and Sign-offs, the accountable owner could be a group of stakeholders, including the stage owners.

Responsibilities

Accountable owners are responsible for reviewing the evidence provided in the checklist and provide approvals required for steps. For items marked as “not applicable” or “not completed,” the accountable owner is responsible for performing risk assessments and approving the continuation with these gaps.

Wider Stakeholders

Role

Wider Stakeholders may include all the parties that play a role relevant to the activities in the step. For example, Data Stewards, Business Stakeholders, MLOps / Data Enablement, Business Impact Monitoring Team.

Responsibilities

Wider stakeholders are engaged in each step for consultation, insights, feedback and/or approval on the process and outcomes and alignment with overall business objectives and strategic goals. These stakeholders should be apprised of the QA and Stage gate and sign-off decisions.

Changing Owners

Owners between steps and stages may change over time, the current and the new owners must ensure handovers are conducted and the changes are tracked over time through, e.g. following the joiner/leaver/mover DX process.



Roles and Responsibilities: Stage 1 – POV (1/2)

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

	1a Use Case Shaping	1b Data Exploration & Preparation	1c Exploratory Data Analysis	1d Feature Engineering	1e Model Training, Evaluation & Validation
Roles and Responsibilities	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - Consult with MLE (if needed) to discuss solution plans to align with MLOps framework and coordinate Data Stewards for data dependencies. - Share the documentations with Business Stakeholder for approval and DX for visibility. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - Collaborate with data stewards to ensure any new data (if applicable) is manipulated adhering to DX principles. - Share data requirements with MLE for data access in deployment framework (if needed). Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - Consult with Business impact monitoring team and Business Owners to ensure that the EDA results align with business needs and objectives. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links are provided.¹ - Consult with Data Stewards to ensure features being stored adheres to DX principles. - Share stored features and documentation with MLE for reproducibility. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links are provided.¹ - Engage with Business impact Monitoring team and Business Owners to ensure that the model performance align with business needs and objectives. - Share model documentation with MLE for reproducibility Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.
Example	Responsible: Data Scientist Accountable: Senior Data Scientist² Wider teams: Data Steward/Data Custodian, MLE, Business Owner, DX	Responsible: Data Scientist Accountable: Senior Data Scientist² Wider teams: DX, MLE/Data Enablement, Business impact Monitoring team	Responsible: Data Scientist Accountable: Senior Data Scientist² Wider teams: Business impact Monitoring team, Business Owner	Responsible: Data Scientist Accountable: Senior Data Scientist² Wider teams: DX, MLE/Data Enablement	Responsible: Data Scientist Accountable: Senior Data Scientist² Wider teams: Business Owner, MLE/Data Enablement

¹ Should an item on the checklist be irrelevant to use case, a justification must be submitted and approved by accountable reviewer to skip the item.

² Accountable owner is one level above Responsible owner.



Roles and Responsibilities: Stage 1 – POV (2/2)

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

Roles and Responsibilities

Example

QA and Stage Gate Review

Responsible Owners

- Ensures all activities and outputs from Stage 1 are reviewed against the established DS Lifecycle framework criteria.
- Conducts a thorough quality assurance review to verify that all steps have been completed and there is clear alignment with the business objectives defined at the outset.
- Coordinate with all stakeholders, including business and technical teams, to finalise the Stage 1 deliverables.

Accountable Owners:

- Confirms that all checklist items are marked as complete, with supporting evidence linked or attached, and the team is ready for a formal sign-off to move to the next stage.

Responsible:

Data Scientist, Senior Data Scientist (to support)

Accountable:

DS Lead, HoD

Wider teams:

Business Owner, Business Impact Monitoring team, DX

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² Accountable owner is one level above Responsible owner.

Stage 1 POV Sign-off

Responsible Owners

- Facilitate formal sign-off meeting where accountable owners can review the outputs of Stage 1 and approve the transition to the next stage.
- Confirm the next steps with accountable owners.
- Collaborate with MLE to handover implementation details for deployment if needed.
- Share the documentations with DX and wider teams for visibility and future reusability.

Accountable Owners

- Providing the relevant approvals to ensure the sign-off process is completed according to the DS Lifecycle framework and risk accepted items are reviewed.

Responsible:

Data Scientist, Senior Data Scientist (to support)

Accountable:

DS Lead, HoD, Business Owner

Wider teams:

MLE/Data Enablement, DX





Roles and Responsibilities: Stage 2 – Productionise (1/2)

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

	2a Model Architecture Refinement	2b Data Integration	2c Model Hardening & Pipeline Creation	2d Model Deployment & Pipeline Management	2e Model Output & Business Integration
Roles and Responsibilities	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - If a handover occurred, ensure communication channels are open with the previous team to align on implementation details. - Coordinate with Data Stewards for data dependencies. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - If a handover occurred, ensure communication channels are open with the previous team to align on implementation details. - Coordinate with Data Stewards for data dependencies. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided.¹ - Consult with Data Scientist to ensure solution design is as expected. - Share implementation plan with Business Owner for visibility. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links are provided.¹ - Facilitate formal deployment meeting between all relevant stakeholders to ensure solution meets expectation and approvals are obtained for deployment. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links are provided.¹ - Consult with Business Owner and Business Impact Monitoring team to ensure model outputs align with expectation. - Collaborate with DX to ensure model outputs are being stored adhering to DX principles. Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.
Example	Responsible: ML Engineer / Data Scientist Accountable: Senior ML Engineer / Senior Data Scientist² Wider teams: Data Steward / Data Custodian, DS, DX	Responsible: ML Engineer / Data Scientist Accountable: Senior ML Engineer / Senior Data Scientist² Wider teams: Data Steward / Data Custodian, DS, DX	Responsible: ML Engineer / Data Scientist Accountable: Senior ML Engineer / Senior Data Scientist² Wider teams: DS, Business Owner	Responsible: ML Engineer / Data Scientist Accountable: Senior ML Engineer / Senior Data Scientist² Wider teams: DS Lead, Business Owner	Responsible: MLE/Data Scientist Accountable: Senior MLE/Senior Data Scientist² Wider teams: Business Owner, DX, Business Impact Monitoring team

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Roles and Responsibilities: Stage 2 – Productionise (2/2)

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

Roles and Responsibilities

Example

QA and Stage Gate Review

Responsible Owners

- Ensure that all activities and outputs from Stage 1 are reviewed against the established DS Lifecycle framework criteria.
- Conduct a thorough quality assurance review to verify that all steps have been completed satisfactorily, and that there is clear alignment with the business objectives defined at the outset.
- Share documentation on deployments and review process , also identifies any gaps, risks, or issues in the model architecture, data integration, and pipeline management.

Accountable Owners

- Oversees the entire QA and Stage Gate Review process, ensuring all checklist items are marked as complete, with supporting evidence linked or attached, and that the team is ready for a formal sign-off by senior leadership.

Responsible: Data Scientist, Senior Data Scientist (to support)
Accountable: DS Lead, HoD
Wider teams : DS, Business Impact Monitoring team, DX

Stage 2 Productionise Sign-off

Responsible Owners

- Facilitate a formal sign-off meeting where accountable owners can review the outputs of Stage 2 and approve ongoing activities.
- Confirm the next steps and ownership with accountable owners.
- Share the relevant documentations with DX and wider teams for visibility and future reusability.

Accountable Owners

- Confirms that the sign-off process is completed according to the DS Lifecycle framework.

Responsible: Data Scientist / ML Engineer, Senior Data Scientist / Senior ML Engineer (to support)
Accountable: DS Lead, HoD, Business Owner
Wider teams : DX, Business Impact Monitoring team

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Roles and Responsibilities: Stage 3 – Operationalise

Our recommendations on the roles and responsibilities required to support the DS Lifecycle.

	3a Model Monitoring	3b Feedback & Reporting	3c QA	Continuous Improvement & Development	Deprecation
Roles and Responsibilities	Responsible Owners - Ensure regular reviews of the deployed model performance has been conducted and the checklist is ticked off and the evidence links have been provided. ¹ - Consult with MLE (if needed) to ensure monitoring systems integrated in production environment and are working correctly. Accountable Owners - Oversees model monitoring process, and all monitoring activities adhere to the framework's checklist and if needed, any issues identified are escalated and addressed promptly.	Responsible Owners - Ensure regular reviews of the business impact has been conducted and the checklist is ticked off and the evidence links have been provided. ¹ - Collaborate with business owner to get KPI's on business value. Accountable Owners - Validate the review process, ensuring any issues identified are escalated and addressed promptly and all insights are communicated effectively to stakeholders.	Responsible Owners - Ensure the checklist is ticked off and the evidence links have been provided. ¹ - Facilitate the QA review process to identify next steps based on monitoring and reporting outcomes. - Collaborate with Business Owner, MLE and Data Science team based on QA outcome. Accountable Owners - Reviews the checklist, ensure all quality checks are completed, share the QA outcome with relevant parties for visibility and approvals.	Responsible Owners - Based on the QA outcome, ensure the checklist is ticked off and the evidence links are provided before moving to the agreed next step. ¹ Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.	Responsible Owners - Ensure the checklist is ticked off and the evidence links are provided. ¹ - Collaborate with DX for retention and deletion policies for deprecation. - Notify the relevant teams and share relevant documentations for visibility Accountable Owners - Reviews the checklist, ensure all relevant approvals are obtained and evidence provided at the end the step as an informal review.
Example	Responsible: Data Scientist Accountable: Senior Data Scientist ² Wider teams: MLE, Business Owner	Responsible: Data Scientist, Business Impact Monitoring Team / Business Stakeholder Accountable: Senior Data Scientist ² Wider teams: Business Owner	Responsible: Data Scientist, Business Impact Monitoring Team / Business Stakeholder Accountable: DS Lead / HoD, Business Owner Wider teams: MLE, Business Teams, DX	Responsible: Data Scientist Accountable: Senior Data Scientist ² Wider teams: DS Lead, MLE, Business Owner, DX	Responsible: MLE, Data Scientist Accountable: Senior Data Scientist ² Wider teams: Business Owner, DX, Business Impact Monitoring team

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